

SOC Tools Setup

Date:19/11/2025

Objective:

Set up essential tools used by SOC Analysts for network analysis, traffic monitoring, and basic incident response.

1. Environment Setup Decision:

A SOC analyst must be comfortable using both Windows and Linux, but for cybersecurity tools, Linux is the best environment.

Decision:

I will install most tools in Kali Linux (Inside my VirtualBox)

Only Wireshark and Sysinternals Suite will be installed on Windows.

2. Tools Installed (Kali Linux)

- Nmap (Pre - Installed)

Purpose: Network scanning, host discovery and service identification.

Installation: Sudo apt install nmap

```
(kali@kali)-[~]
$ nmap --version
Nmap version 7.95 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.7 openssl-3.5.2 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.5 nmap-libnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
```

- Wireshark

Purpose: GUI packet analyzer for inspecting network traffic

Installation: Sudo apt install wireshark

```
(kali@kali)-[~]
$ wireshark --version
Wireshark 4.4.7.

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NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compiled (64-bit) using GCC 14.3.0, with GLib 2.84.4, with Qt 6.8.2, with
libpcap, with POSIX capabilities (Linux), with libnl 3, with zlib 1.3.1, without
zlib-ng, with PCRE2, with Lua 5.4.8, with GnuTLS 3.8.10 and PKCS #11 support,
with Gcrypt 1.11.2, with Kerberos (MIT), with MaxMind, with nghttp2 1.64.0, with
nghttp3 1.8.0, with brotli, with LZ4, with Zstandard, with Snappy, with libxml2
2.14.5, with libbrotli 1.1.0, with minizip 1.3.1, with QtMultimedia, with QtDBus,
without automatic updates, with binary plugins.

Running on linux 6.12.38-kali-amd64, with Intel(R) Core(TM) i7-10610U CPU @
1.80GHz (with SSE4.2), with 1949 MB of physical memory, with GLib 2.84.4, with
Qt 6.8.2, with libpcap 1.10.5 (with TPACKET_V3), with zlib 1.3.1, with PCRE2
10.45 2025-02-05, with c-ares 1.34.5, with GnuTLS 3.8.9, with Gcrypt 1.11.2,
with nghttp2 1.64.0, with nghttp3 1.8.0, with brotli 1.1.0, with LZ4 1.10.0,
with Zstandard 1.5.7, with libbrotli 1.1.0, with LC_TYPE=en_US.UTF-8, binary
plugins supported.
```

- Tshark

Purpose: Terminal version of Wireshark for packet capture and analysis from CLI.

Installation: `sudo apt install tshark`

```
(kali@kali)-[~]
$ tshark --version
Tshark (Wireshark) 4.4.7.

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NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compiled (64-bit) using GCC 14.3.0, with Glib 2.84.4, with libpcap, with POSIX
capabilities (Linux), with libnl 3, with zlib 1.3.1, without zlib-ng, with
PCRE2, with lua 5.4.8, with GnuTLS 3.8.10 and PKCS #11 support, with Gcrypt
1.11.2, with Kerberos (MIT), with MaxMind, with nghttp2 1.64.0, with nghttp3
1.8.0, with brotli, with LZ4, with Zstandard, with Snappy, with libxml2 2.14.5,
with libsmi 0.4.8, with binary plugins.

Running on Linux 6.12.38+kali-amd64, with Intel(R) Core(TM) i7-10610U CPU @
1.80GHz (with SSE4.2), with 1949 MB of physical memory, with Glib 2.84.4, with
libpcap 1.10.5 (with TPACKET_V3), with zlib 1.3.1, with PCRE2 10.43 2023-02-03,
with c-ares 1.34.5, with GnuTLS 3.8.9, with Gcrypt 1.11.2, with nghttp2 1.64.0,
with nghttp3 1.8.0, with brotli 1.1.0, with LZ4 1.10.0, with Zstandard 1.5.7,
with libsmi 0.4.8, with LC_TYPE=en_US.UTF-8, binary plugins supported.
```

- Tcpdump

Purpose: lightweight command - line packet capture and analyzer tool.

Installation: `sudo apt install tcpdump`

```
(kali@kali)-[~]
$ tcpdump --version
tcpdump version 4.99.5
libpcap version 1.10.5 (with TPACKET_V3)
OpenSSL 3.5.2 5 Aug 2025
64-bit build, 64-bit time_t
```

- Netcat(nc)

Purpose: Testing ports, transferring files, creating simple listeners.

Installation: `sudo apt install netcat-traditional`

```
(kali@kali)-[~]
$ nc -h
[vi.10-50]
connect to somewhere: nc [-options] hostname port[s] [ports] ...
listen for inbound: nc -l -p port [-options] [hostname] [port]
options:
-c filename as '-e': use /bin/sh to exec [dangerous!!]
-b allow broadcasts
-k gateway source-routing hop point[s], up to 8
-h num source-routing pointer: 4, 8, 12, ...
-i secs this craft
-l delay interval for lines sent, ports scanned
-k listen mode, for inbound connects
-n numeric-only IP addresses, no DNS
-o file hex dump of traffic
-p port local port number
-r randomize local and remote ports
-q secs quit after EOF on stdin and delay of secs
-s addr local source address
-t tos set Type Of Service
-T answer TELNET negotiation
-u UDP mode
-v verbose [use twice to be more verbose]
-w secs timeout for connects and final net reads.
-z Send CR/LF as line-ending
-Z zero-I/O mode [used for scanning]
PORT numbers can be individual or ranges: lo-hi [inclusive]
hyphens in port names must be backslash escaped (e.g., 'ftp\data').
```

- Burpsuite(Pre-Installed)

Purpose: Web traffic interception, penetration testing.

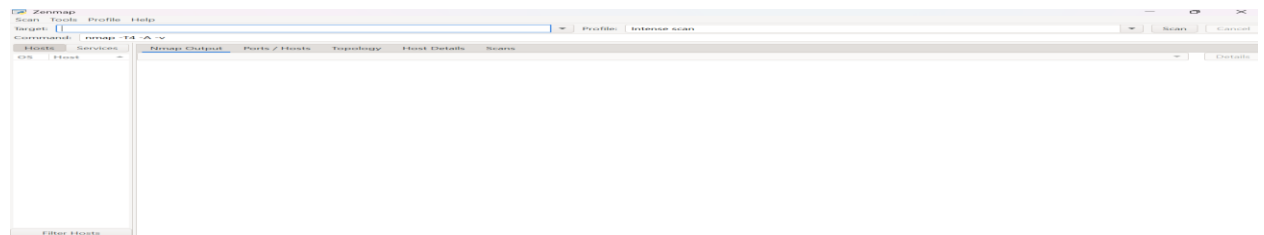
Installation: Already installed in kali Linux (Application > web application analysis > Burp Suite)

3. Tools Installed(Windows)

- Wireshark

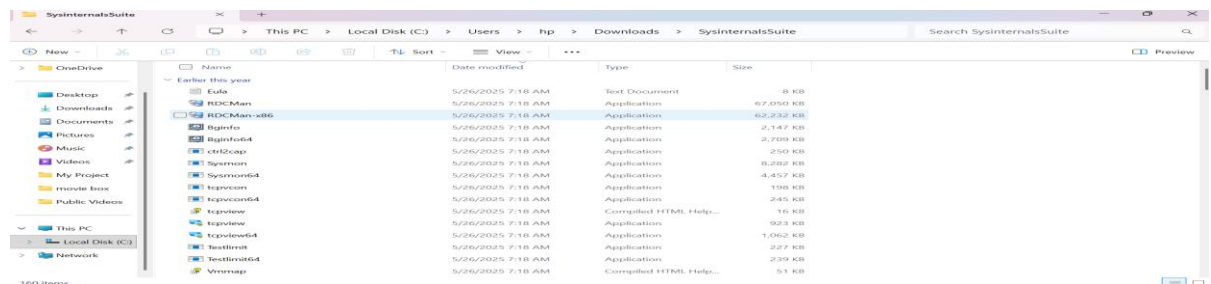


- Nmap (Zenmap GUI)



- Sysinternals Suite

Purpose: It is a collection of advanced utilities for Microsoft Windows, used for deep system troubleshooting, diagnosis, and performance analysis.



4. Summary

Today, I set up the essential tools needed for SOC operations. Most tools are installed in Kali Linux in VirtualBox because it is the most efficient environment for cybersecurity analysis. Windows tools are kept for flexibility.

